TE 357: Optical Communications

Lecture 3: Optical Fibers

Lecture Objectives

1. To gain insight into the propagation of light along an optical fiber
2. To explain some phenomena that light goes through

Lecture Objective 1

- To gain insight into the propagation of light along an optical fiber
- Thus
  - To determine
    - the conditions needed to transmit light through an optical fiber
      - and resolve how these conditions can be implemented in a practical manner

Lecture Objective 2

- To explain some phenomena that light goes through
- Thus
    - Attenuation
        - Definition
        - Mechanisms
        - Loss calculation

Lecture Objective 2

- Thus
  - Dispersion
    - definition
    - Concept of modes
    - Calculation of number of modes
    - Modal (intermodal) dispersion
      - Pulse spread calculation

Introduction

- What is an optical fiber?

Introduction

- What is an optical fiber?
  - It is a thin, transparent, flexible strand
  - It is a dielectric waveguide that operates at optical frequencies

Introduction

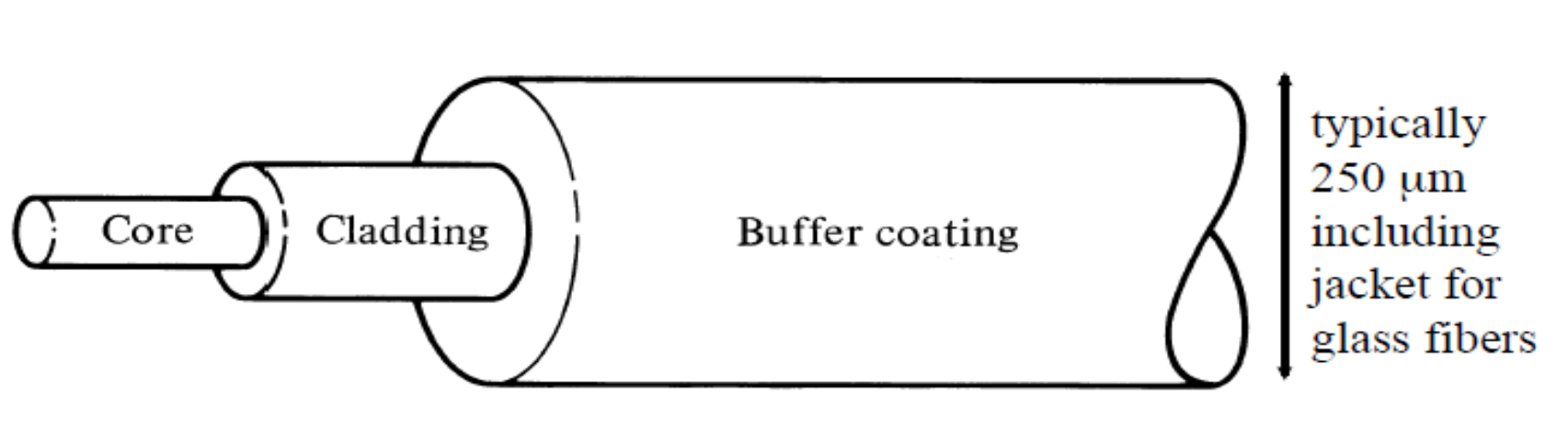
- What is an optical fiber?
  - It is normally cylindrical in form and confines electromagnetic energy in the form of light to within its surfaces.
    - It guides the light in a direction parallel to its axis

Introduction

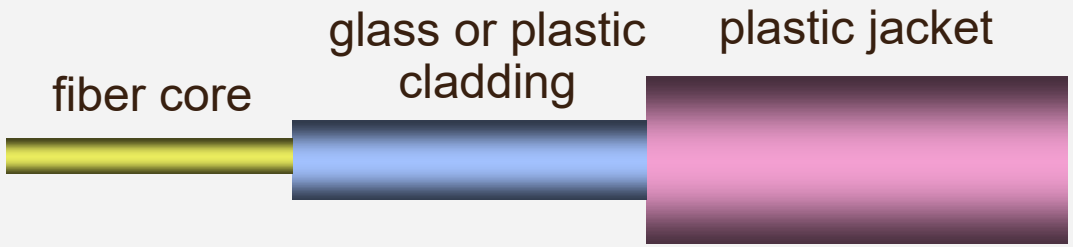
- Some questions that arise concerning optical fibers are:
  1. What is the structure of an optical fiber?
  2. Of what materials are fibers made?
  3. How does light propagate along a fiber?
  4. What is the signal loss or attenuation mechanism in a fiber?
  5. Why and to what degree does a signal get distorted as it travels along a fiber?

Optical Fiber- Basic structure

- Optical fiber structure
  - Core and cladding made from the same material (e.g. silica)
    - differ only in their refractive indexes (why?)

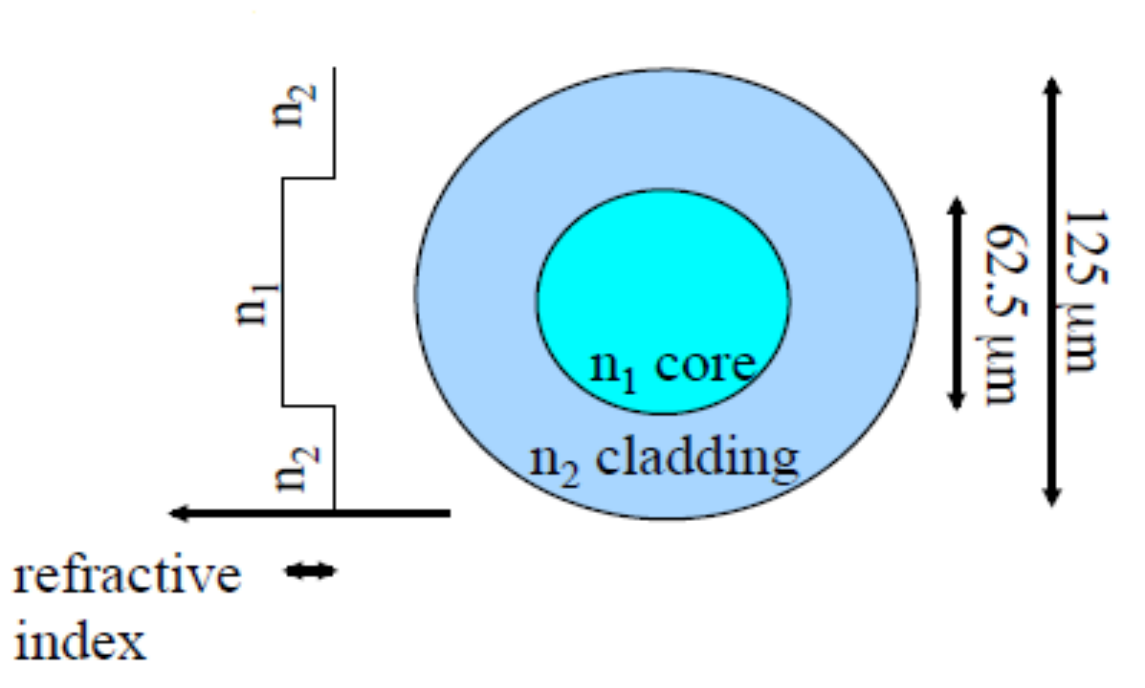


Optical Fiber-Basic structure



Optical Fiber

- Optical fiber structure



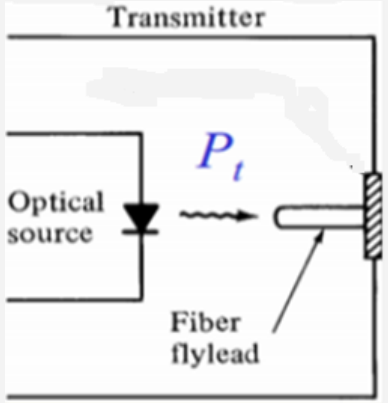
- Step index fiber

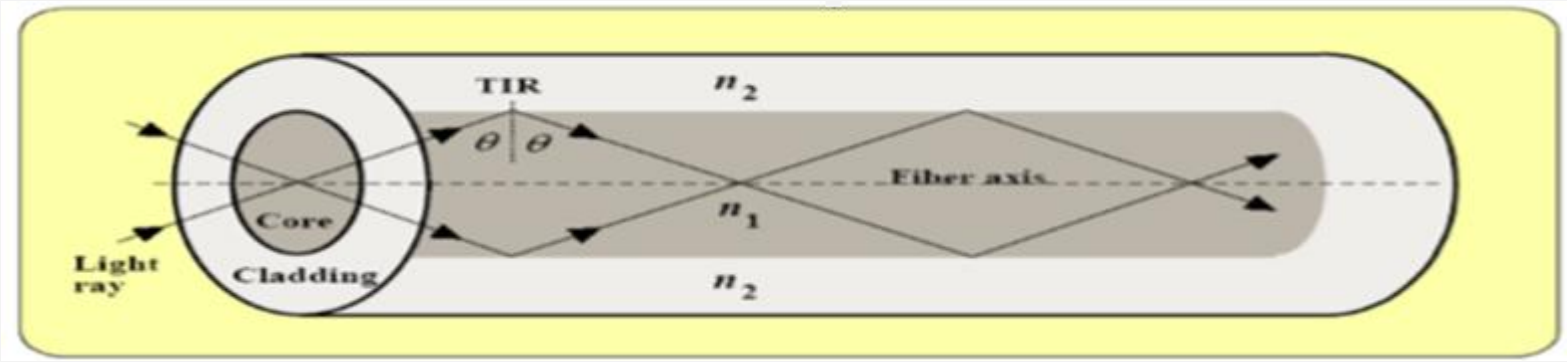
Light ray guiding condition

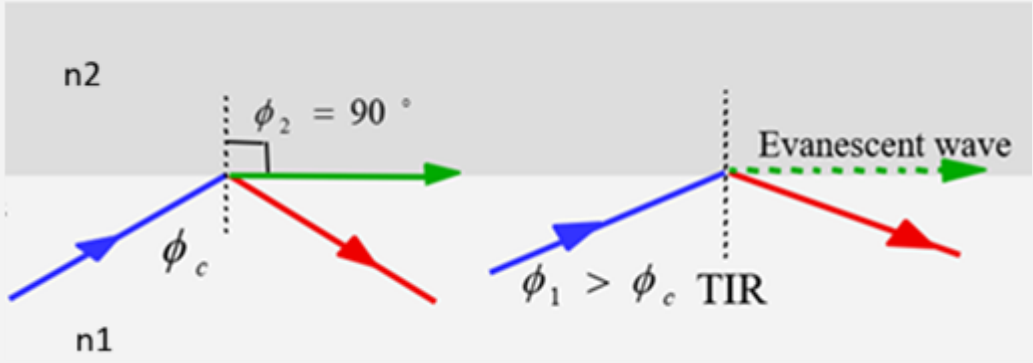
- To save light
  - inside a strand of fiber, what needs to happen?

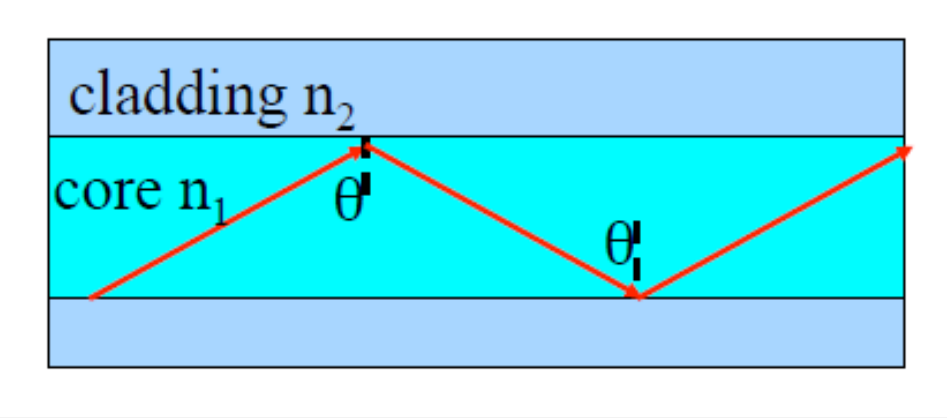
Light ray guiding condition

- Framework



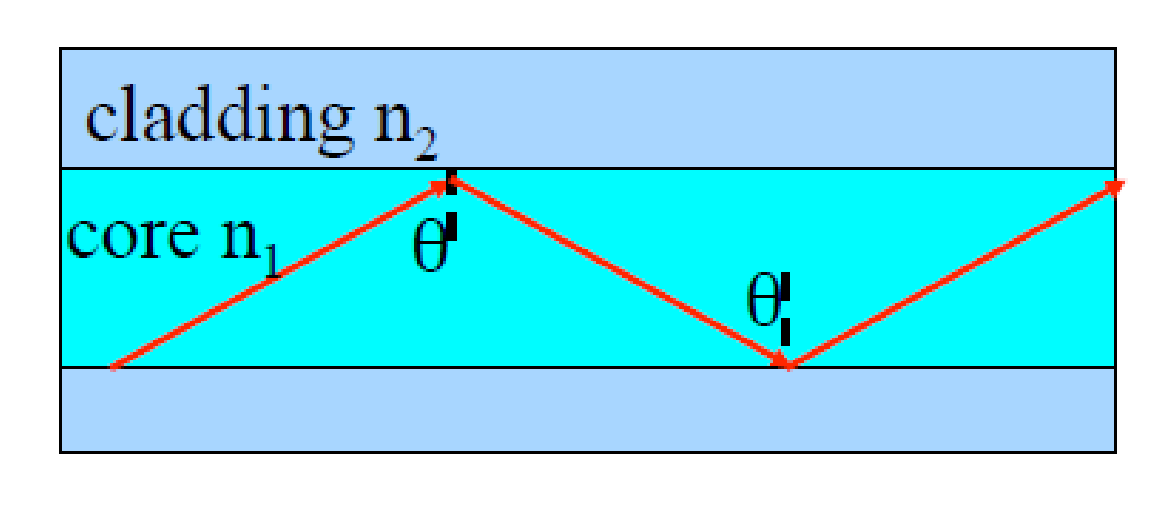






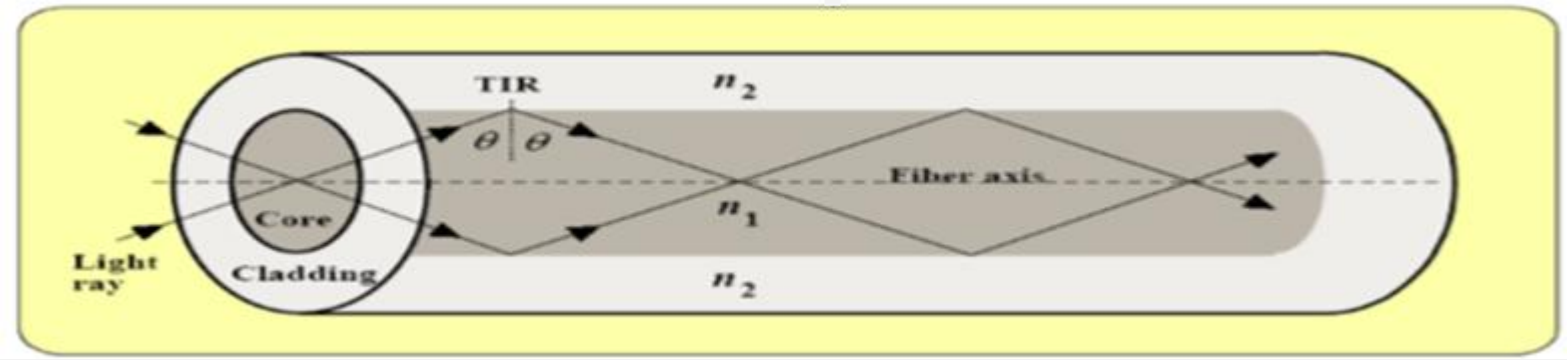
Light ray guiding condition

- To save light
  - inside a strand of fiber, we need to have it strike the
    - core-cladding boundary at an angle above the critical incident angle, θ_c, in order to provide total reflection of this light



Light ray guiding condition

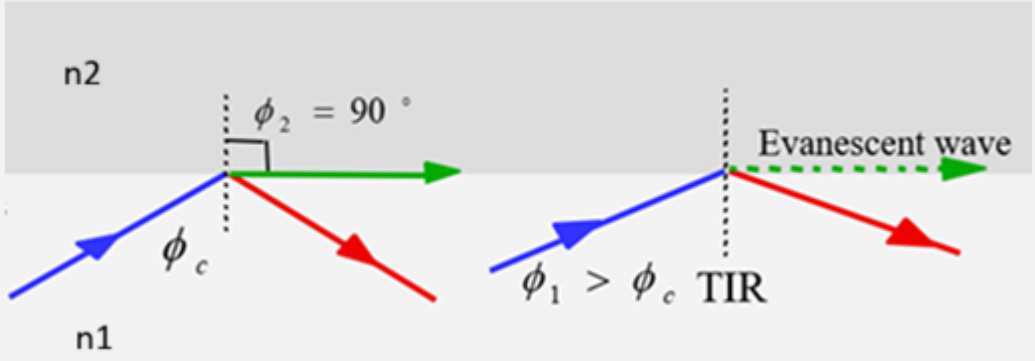
- Light ray that satisfies
  - total internal reflection condition at the Interface/boundary of the CORE and the CLADDING can be guided along an optical fiber



Light ray guiding condition

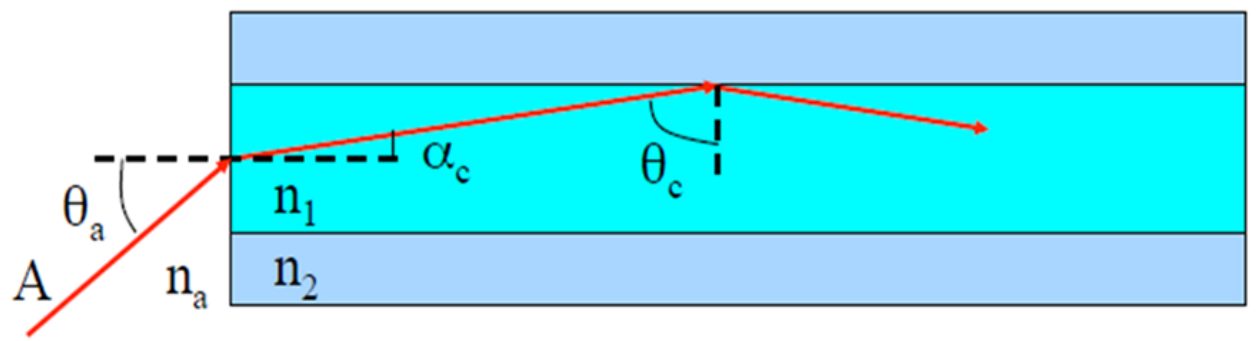
- Under what condition will light be trapped inside the fiber core?
  i. $n_1 = 1.48$ and $n_2 = 1.46$ (silica)
  ii. $n_1 = 1.495$ and $n_2 = 1.402$ (Plastic)

$$\phi_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$



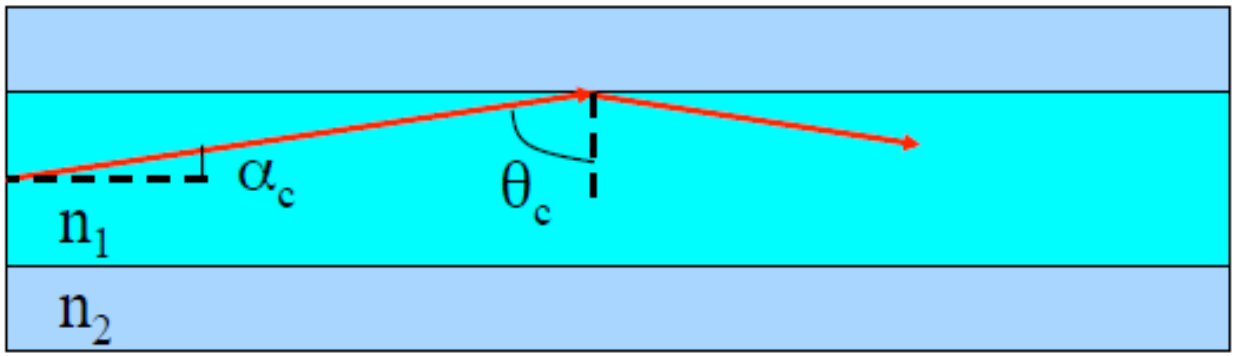
Light ray guiding condition

- Total Internal Reflection (TIR)
  - Critical incident angle
  - Critical propagation angle
- Launching of light
  - Acceptance angle
  - Numerical aperture



Critical- Incident & Propagation angles

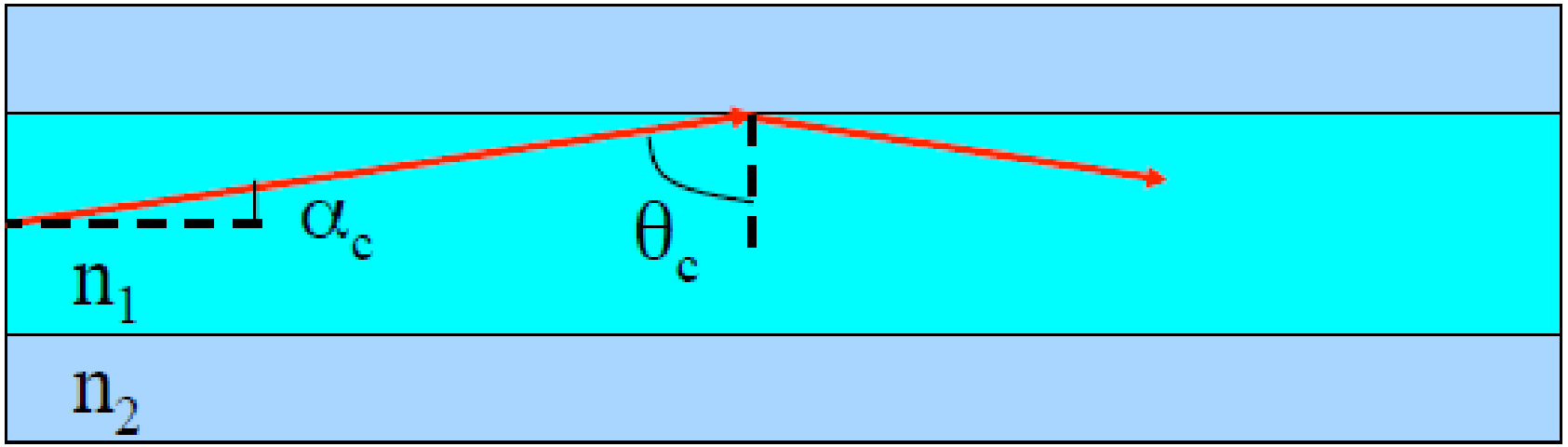
- Two key terms
  i. Critical incident angle, θ_c
     - Is the angle the beam makes with the line perpendicular to the boundary between the core and the cladding
  ii. Critical propagation angle, α_c
     - Is the angle the beam makes with the centerline of the optical fiber



Critical- Incident & Propagation angles

- Two key terms
  - Critical incident angle, θ_c
  - Critical propagation angle, α_c

$$\alpha_c = 90 - \theta_c$$



Critical- Incident & Propagation angles

- We know that

$$\sin\theta_c = \frac{n_2}{n_1}$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

- Since

$$\alpha_c = 90 - \theta_c$$

- It implies

$$\cos\alpha_c = \sin\theta_c$$

Critical- Incident & Propagation angles

- thus

$$\sin^2 x + \cos^2 x = 1$$

$$\cos\alpha_c = \frac{n_2}{n_1}$$

$$\sin\alpha_c = \sqrt{1-\left(\frac{n_2}{n_1}\right)^2}$$

$$\alpha_c = \sin^{-1}\sqrt{\left(1-\left(\frac{n_2}{n_1}\right)^2\right)}$$

Critical- Incident & Propagation angles

- The critical propagation angle α_c
  - represents the requirement of achieving total internal reflection
- What is the critical propagation angle α_c
  - For $n_1 = 1.48$ and $n_2 = 1.46$
- State the requirement
  - of achieving total internal reflection in terms of critical propagation angle α_c

Critical- Incident & Propagation angles

- For n_1 = 1.48 and n_2 = 1.46

$$\alpha_c = \sin^{-1}\sqrt{\left(1-\left(\frac{1.46}{1.48}\right)^2\right)} = 9.43^\circ$$

- What happens
  - when a beam travels at a propagation angle of 10 degrees?

Critical- Incident & Propagation angles

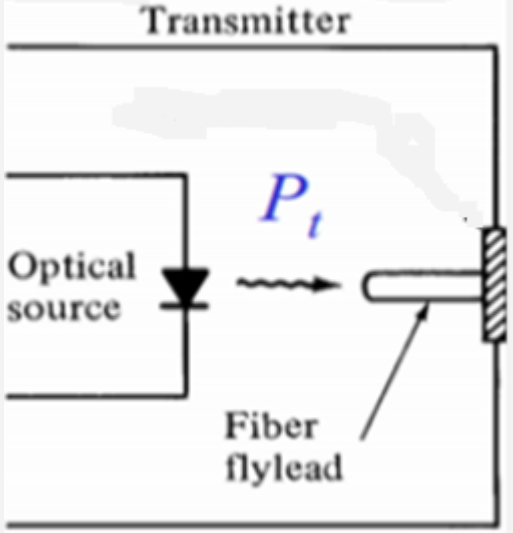
- In conclusion,
  - To save light
    - inside a strand of fiber, we need to have it strike the
      - core-cladding boundary at an angle above the critical incident angle, θ_c, in order to provide total reflection of this light
        - To make light fall above that angle, we have to direct it so that it is below the critical propagation angle, α_c ,with respect to the centerline of the fiber

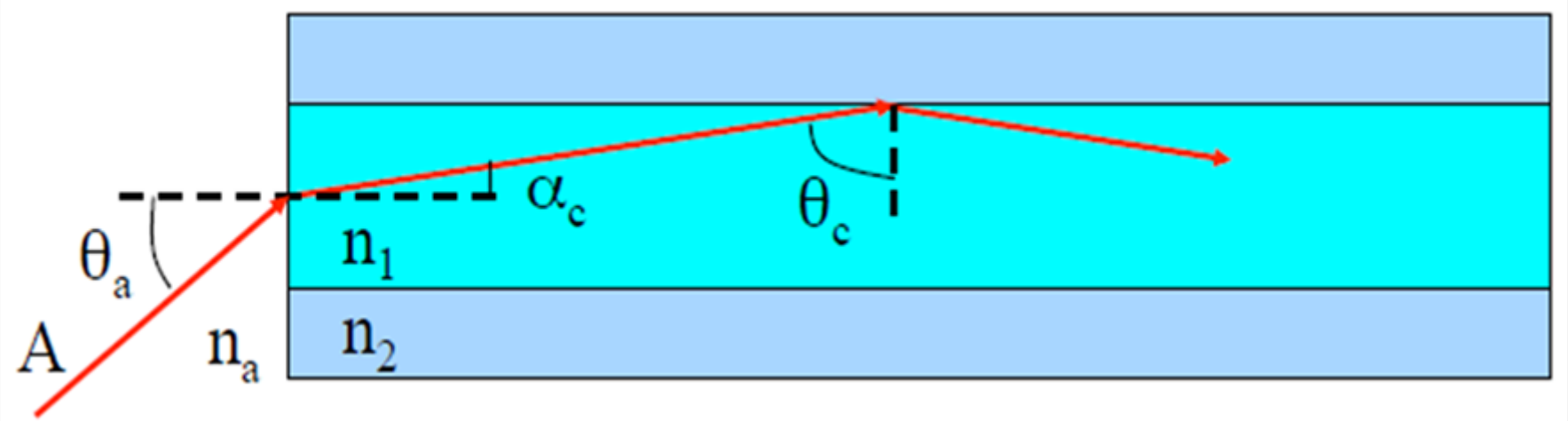
Acceptance Angle

- The next question that arises is
  - How can we direct this beam so that
    - it does indeed fall below the critical propagation angle?

Acceptance Angle

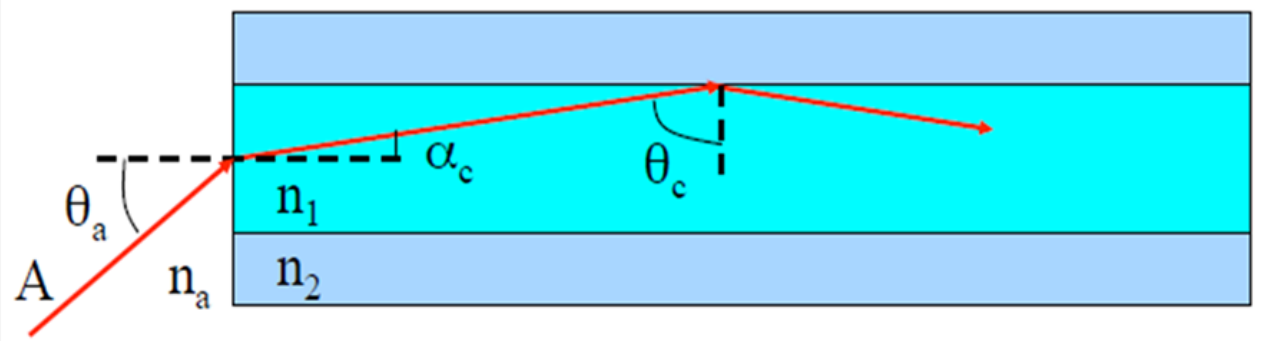
- The light
  - of course, must come from a source that is outside the fiber and therefore have to be directed into the fiber





Recap

- Light ray guiding condition
  - Total Internal Reflection (TIR)
    - Critical incident angle
    - Critical propagation angle
  - Launching of light
    - Acceptance angle
    - Numerical aperture

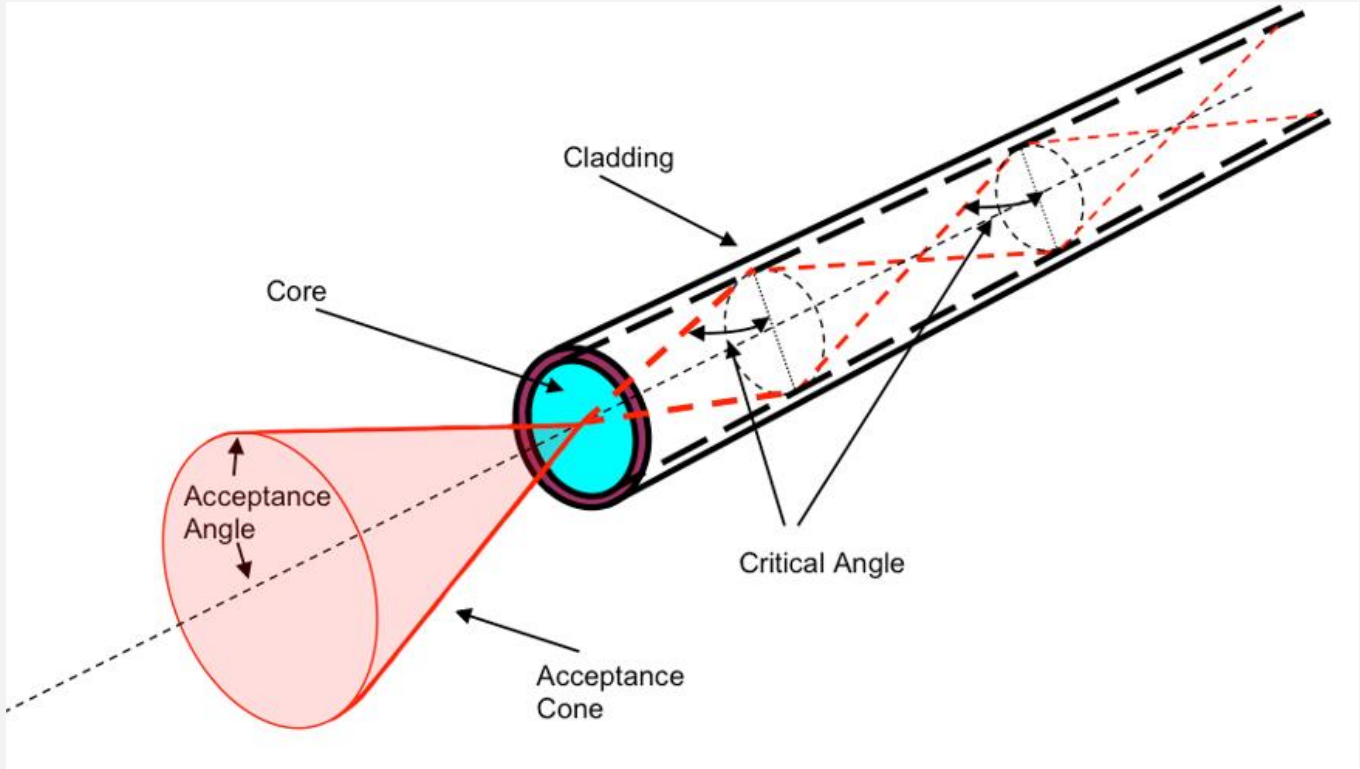


Acceptance Angle

- To save light
  - inside a fiber all rays must propagate at an angle less than the critical propagation angle α_c
    - In order for light to be maintained in the fiber at this angle, it must be directed from outside the fiber at angle θ_a or less

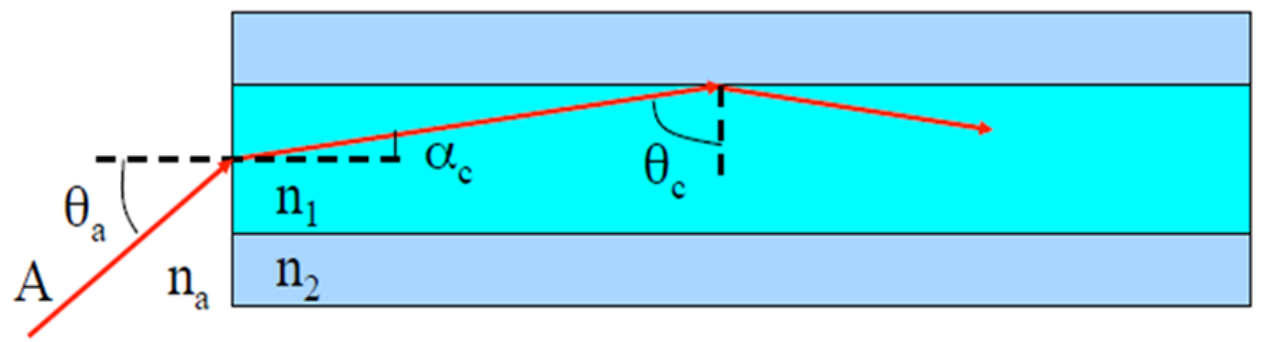
Acceptance Angle

- Light will be
  - saved inside the fiber if it comes from a source bounded by the cone $2\theta_a$ (cone of light)



Acceptance Angle

- Launching light into an optical fiber
  - at the gap-fiber interface, the beam at angle θ_a is the incident beam and the beam at angle α_c is the launched one

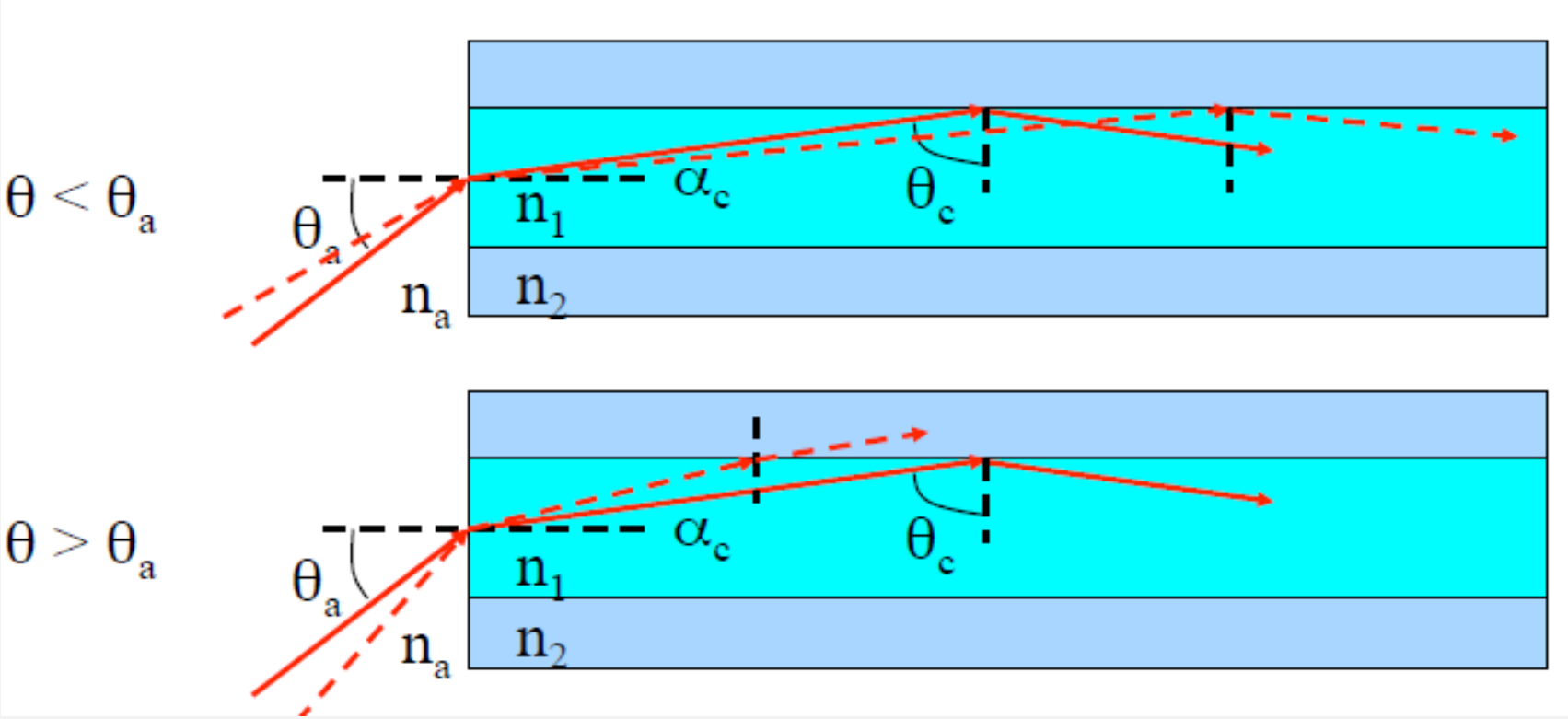


- The relationship between
  - θ_a and α_c can be derived using Snell's law

$$n_a \sin\theta_a = n_1 \sin\alpha_c$$

Acceptance Angle

- What happens when the incidence beam, θ is less than θ_a ?
- What happens when the incidence beam, θ is greater than θ_a ?

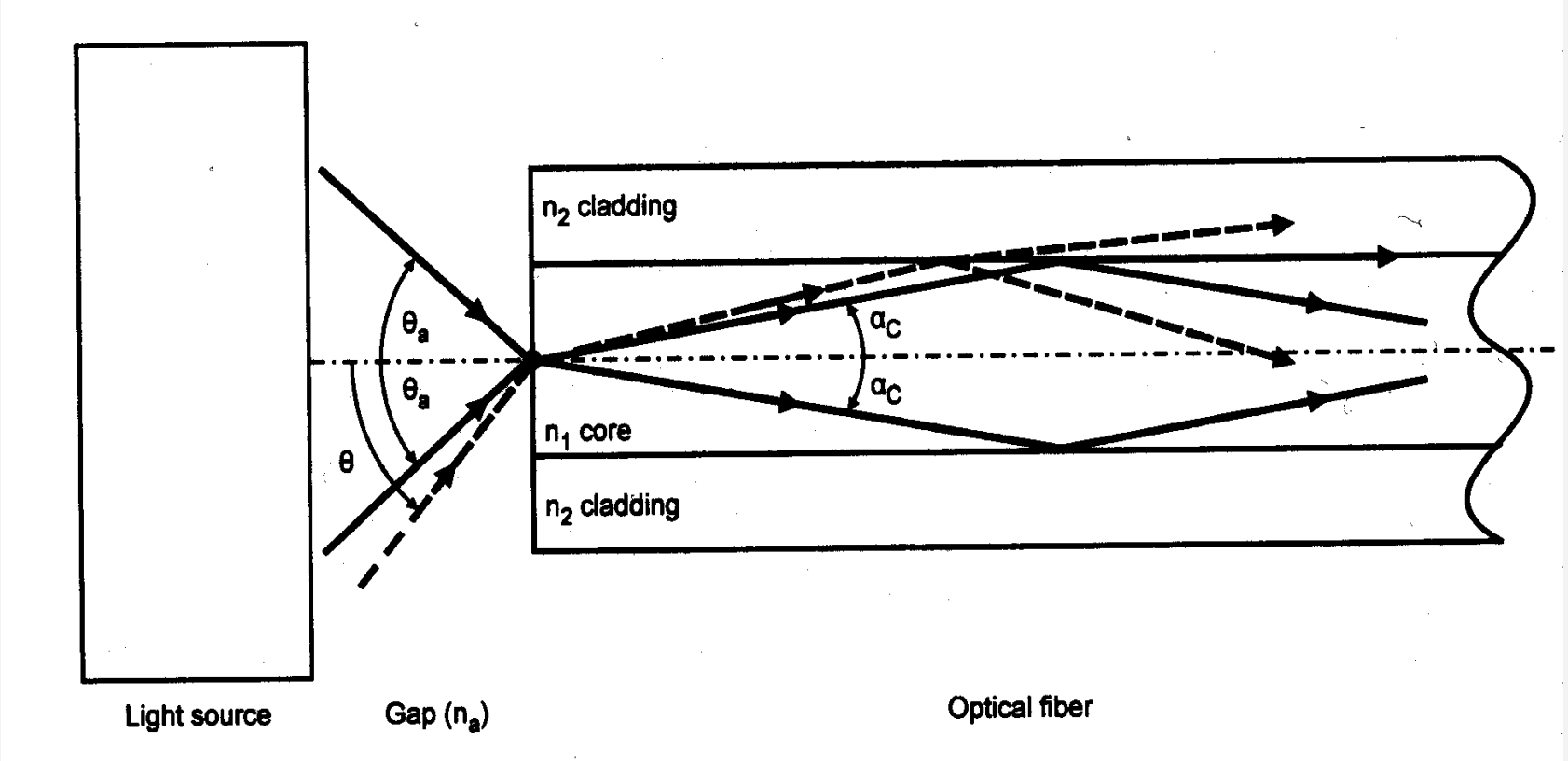


Acceptance Angle

- Any ray which is incident into the fiber core at
  - an angle greater than θ_a will have a propagation angle greater than α_c and therefore incident angle less than θ_c at the core-cladding interface and will NOT be totally internal reflected

Acceptance Angle

- Thus θ_a defines
  - an acceptance cone for an optical fiber



Acceptance Angle

- Light will be
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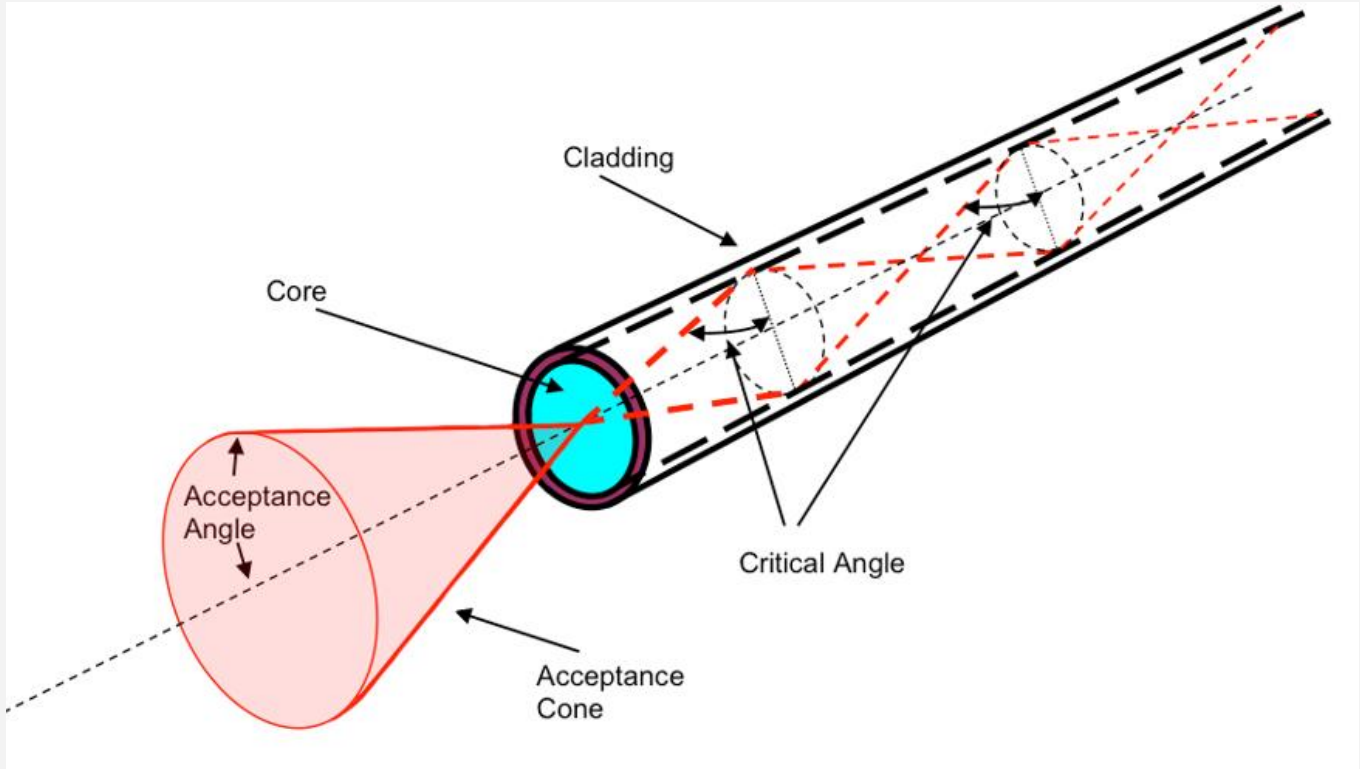


Acceptance Angle

- To save light
  - inside a fiber all rays must propagate at an angle less than the critical propagation angle α_c
    - In order for light to be maintained in the fiber at this angle, it must be directed from outside the fiber at angle θ_a or less

Acceptance Angle

- What
  - is the acceptance angle and cone of light for a fiber when $n_1 = 1.48$ and $n_2 = 1.46$

Acceptance Angle

- What
  - is the acceptance angle and cone of light for a fiber when $n_1 = 1.48$ and $n_2 = 1.46$

$$\sin\theta_a = n_1 \sin\alpha_c$$

$$\sin\alpha_c = \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

Acceptance Angle

- The acceptance angle is
  - described using Numerical Aperture (parameter in data sheets)

Numerical Aperture (NA)

- NA describes
  - the ability of an optical fiber to gather light from a source and
    - the ability to preserve or save, this light inside the fiber because of total internal reflection

$$NA = \sin\theta_a$$

Numerical Aperture (NA)

- We know that

$$\sin\theta_a = n_1 \sin\alpha_c$$

- But

$$\sin\alpha_c = \sqrt{1-\left(\frac{n_2}{n_1}\right)^2}$$

- therefore

$$NA = n_1 \sin\alpha_c = n_1\sqrt{1-\left(\frac{n_2}{n_1}\right)^2} = \sqrt{(n_1)^2-(n_2)^2}$$

Numerical Aperture (NA)

- It is very common
  - to use the relative difference of the refractive indexes, Δ, often called the relative index
- Thus

$$\Delta = \frac{n_1 - n_2}{n_1}$$

- We assume

$$n_1 \approx \frac{n_1 + n_2}{2}$$

Numerical Aperture (NA)

- The numerical aperture

$$NA = \sqrt{(n_1)^2 - (n_2)^2} = \sqrt{(n_1 - n_2)(n_1 + n_2)}$$

- But

$$(n_1 - n_2) = n_1 \Delta$$

$$(n_1 + n_2) = 2 n_1$$

- thus

$$NA \approx \sqrt{n_1 \Delta \cdot 2 n_1} \approx n_1 \sqrt{2\Delta}$$

Problem

- Calculate the numerical aperture for the following two fibers
  i. $n_1 = 1.48$ and $n_2 = 1.46$ (silica)
  ii. $n_1 = 1.495$ and $n_2 = 1.402$ (Plastic)

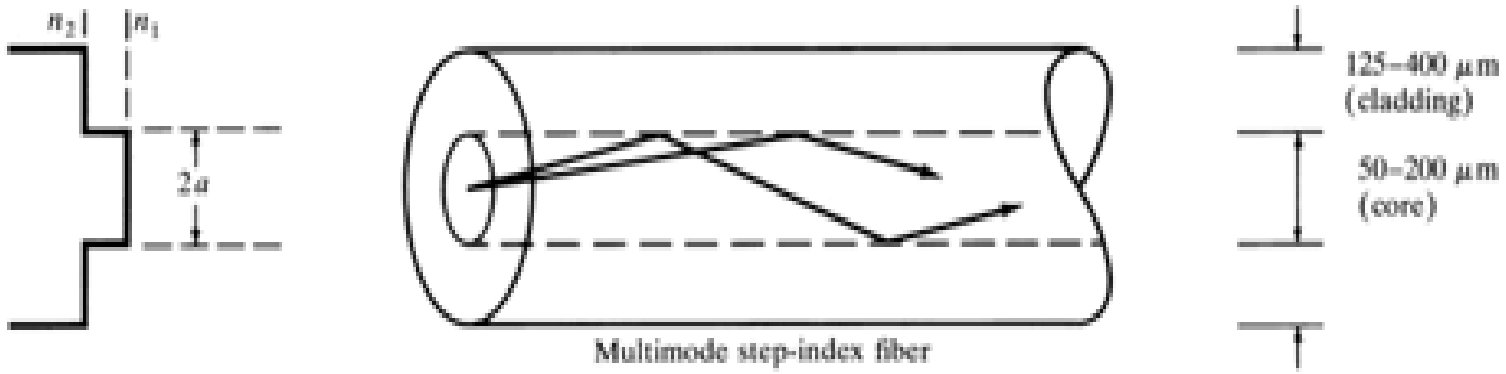
- Comment on your answers

Problem

- Briefly explain the phenomenon called total internal reflection and its importance in fiber optic communication technology
- Explain what is meant by the following terms and their importance in fiber-optic communication technology
  - Critical incident angle
  - Critical propagation angle
  - Acceptance angle
  - Numerical aperture
- For specific fiber, NA = 0.275 and n_1= 1.49
  - Find the critical propagation angle

Type of Fiber

- So far
  - The basic type of fiber optic we encounter is:
  1. Step-index multimode



Multimode step-index fiber

Focus

- Objectives
  1. To gain insight into the propagation of light along an optical fiber
  2. To explain some phenomena that light goes through
     a. Attenuation (Reduction in power level with distance)
        - Definition
        - Mechanisms
        - Loss calculation
     b. Dispersion (eroding of clarity with distance and speed)
        - definition
        - Concept of modes
        - Calculation of number of modes
        - Multimode fiber
          - Modal (intermodal) dispersion (cause)
            - Pulse spread calculation

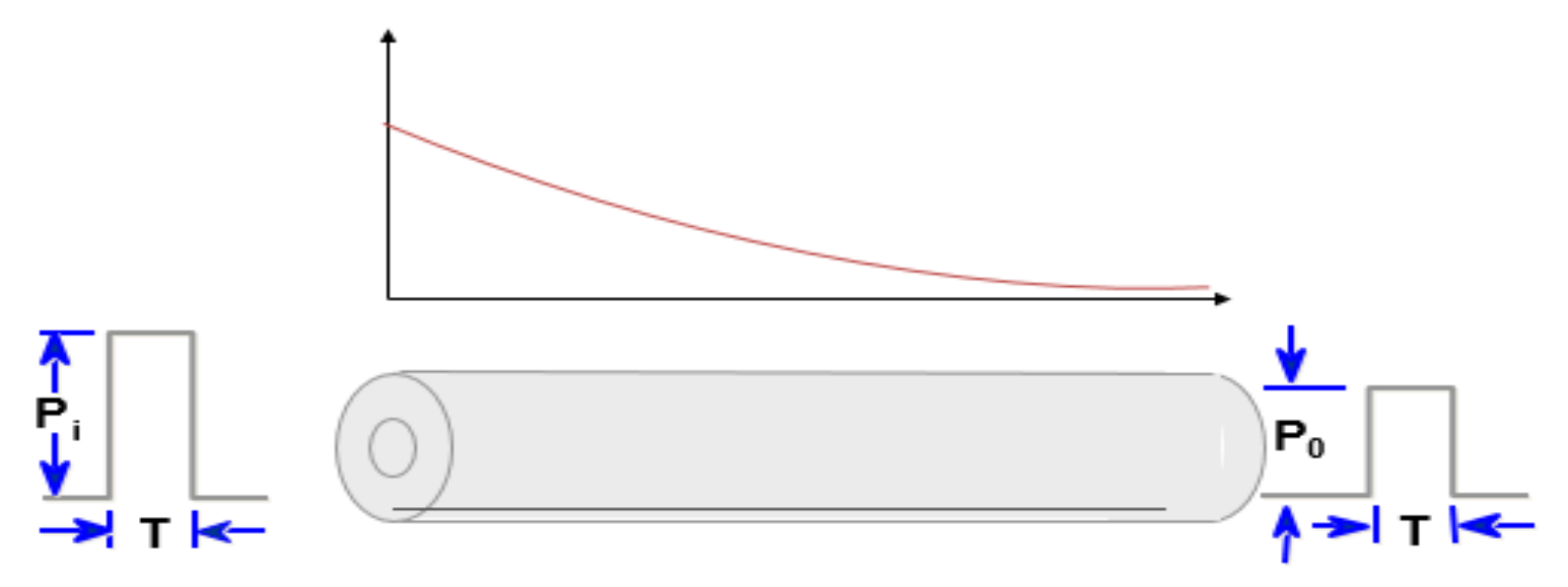
Attenuation and Dispersion

- This section
  - describes the operational characteristics of fibers
- Thus
  1. What are the loss or signal attenuation mechanisms in a fiber?

  2. Why and to what degree do optical signals get distorted as they propagate along a fiber?

Attenuation - Definition

- Signal attenuation
  - In fiber-optic communications
    - attenuation is the decrease in light power during light propagation along an optical fiber



Attenuation - Definition

- Signal attenuation
  - is one of the most important properties of an optical fiber
    - Because it largely determines the maximum unamplified or repeater-less separation between a transmitter and a receiver

Attenuation - Mechanisms

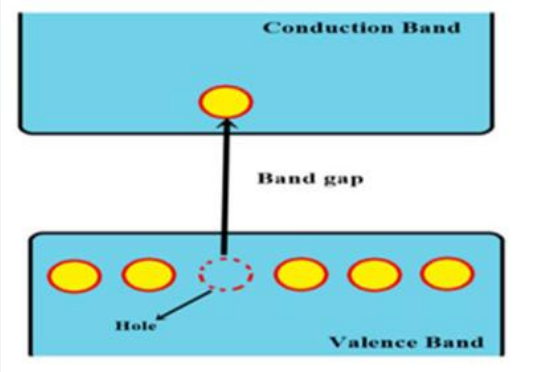
- The basic attenuation mechanisms in a fiber are
  - Intrinsic losses
    - i. Absorption
    - ii. Scattering
  - Extrinsic losses
    - Bending (micro-bending and macro-bending)

Mechanisms - Absorption

- Recall that
  - If an incoming photon has a frequency, f, wavelength λ, such that its energy,

$$E_P = hf = \frac{hc}{\lambda}$$

  - is equal to the energy gap of the material, the photon will be absorbed by the material



Mechanisms - Absorption

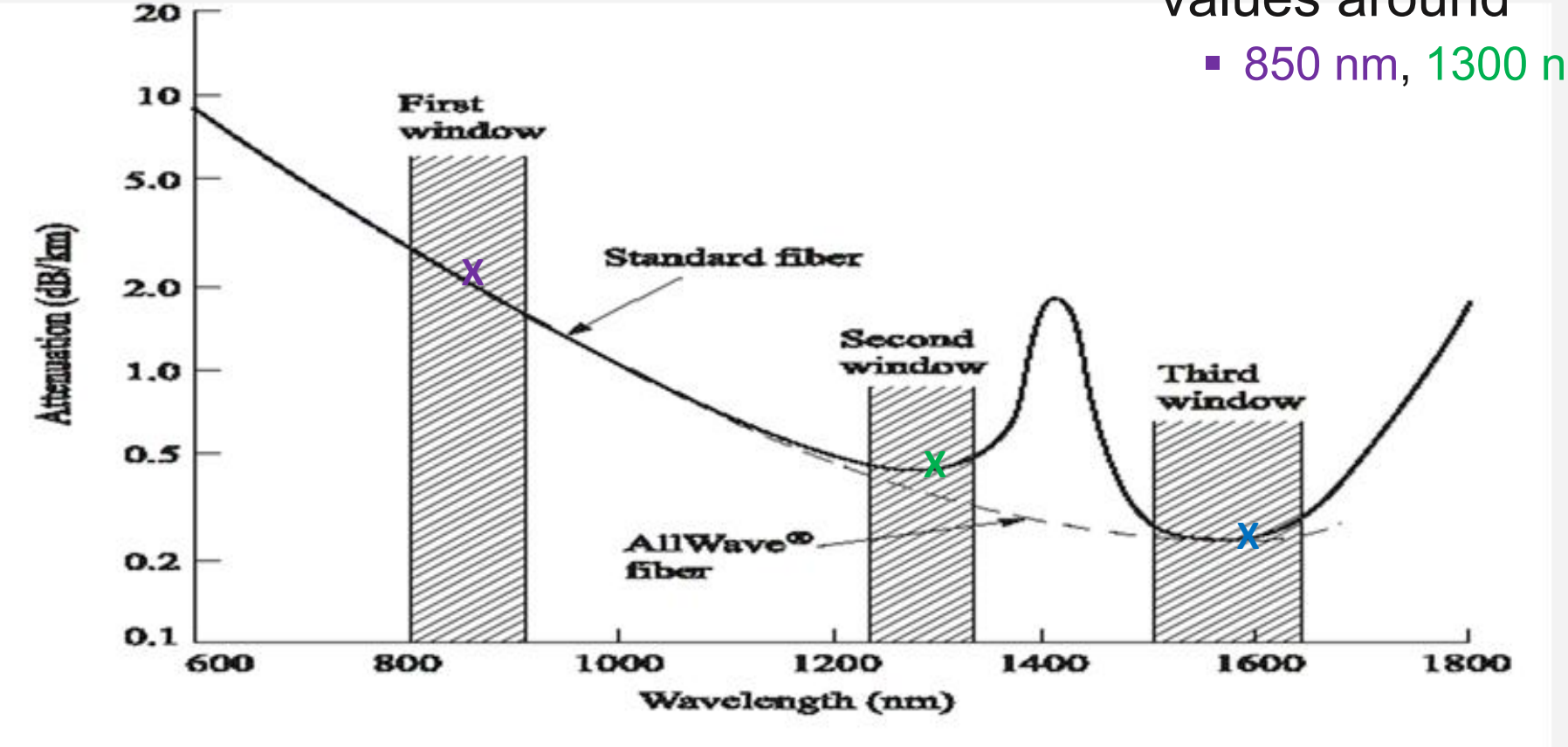
- When
  - light travels down an optical fiber and encounters a material whose energy gap is exactly equal to the energy of
    - these photons, the light will be absorbed
      - This type of loss is wavelength dependent

Mechanisms - Absorption

- Optical fiber
  - is a transparent strand, that is ‘non absorptive’ material
    - Absorption properties that still remain are caused not by silica atoms but by some molecules of the hydroxide anion OH^-
      - These molecules are incorporated in silica during the fabrication process and is very hard to eliminate them
- OH^- molecules have major peaks of absorption at
  - 945 nm, 1240 nm, and 1380 nm

Spectral – Attenuation Curve

- Attenuation reaches minimum values around
  - 850 nm, 1300 nm, and 1550 nm



Transmission Spectral (Transparent) Windows

- Three major fiber transmission spectral windows

The 1st window: 850 nm, attenuation 4 dB/km

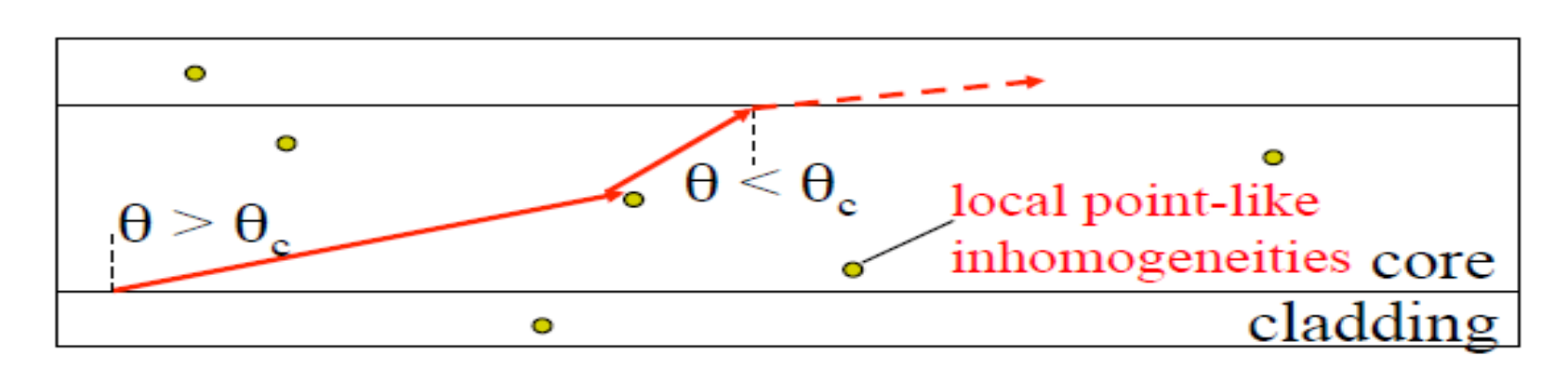
The 2nd window: 1300 nm, attenuation 0.5 dB/km

The 3rd window: 1550 nm, attenuation 0.2 dB/km

1550 nm window is today's standard **long-haul** communication wavelengths.

Mechanism - Scattering Loss

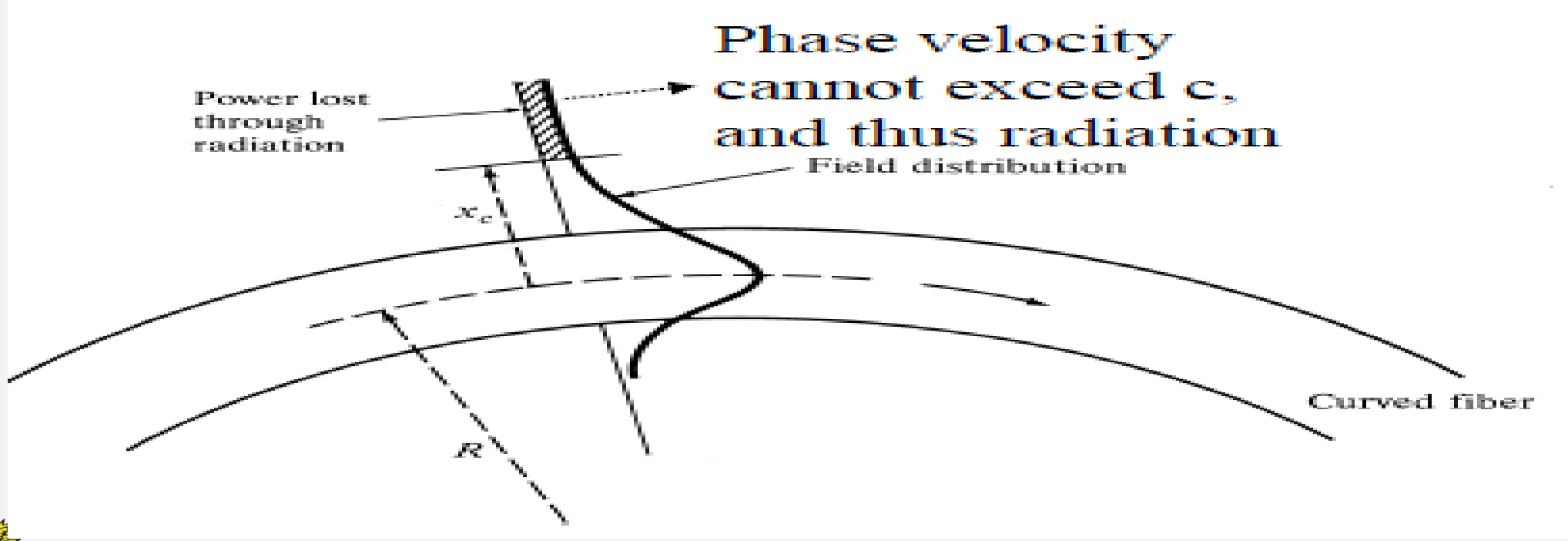
- When there is an imperfection in the core material
  - a beam propagating at the critical propagation angle or less changes direction after it meets an obstacle
    - That is light will be scattered



    - Imperfection means
      - slight variations in the refractive index

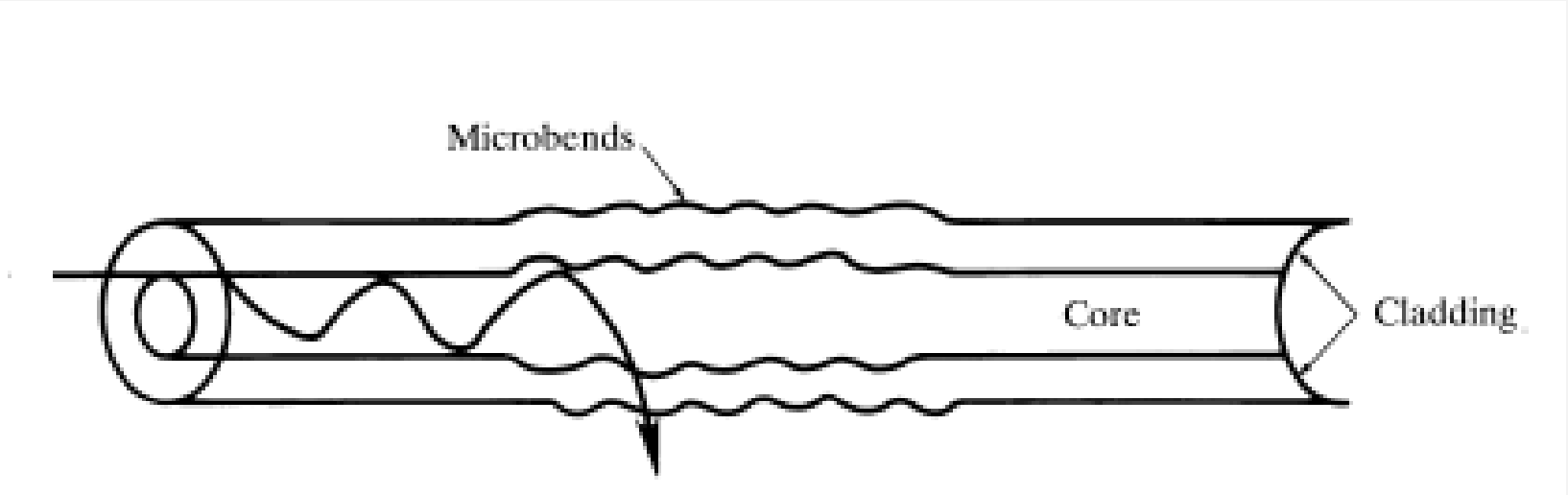
Mechanism - Bending loss

- Macro-bending
  - Loss caused by the curvature of the entire axis



Mechanism - Bending loss

- Micro-bending
  - Loss caused by micro-deformations of the fiber axis



- Bending of the fiber
  - leads to disruption of the condition of total internal reflection

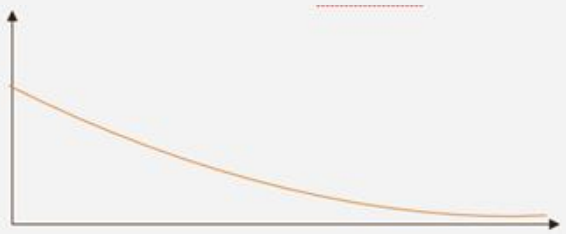
Optical Attenuation

- Signal amplitude reduction limits “how far”
- Attenuation in dB
- Power is measured in dBm:

Calculation of total Attenuation

- Fiber loss is
  - the ratio of power at the output end of fiber, P_{out}, to power launched into the fiber, P_{in}

$$Loss = \frac{P_{out}}{P_{in}}$$



  - where power is measured in watts

Calculation of total Attenuation

- In communication technology
  - Loss (attenuation) is measured in decibels (dB)

$$Loss(dB) = -10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

  - The negative is introduced
    - in order to present loss as a positive quantity

Calculation of total Attenuation

- Attenuation per unit of fiber length, A is

$$A(dB/km) = Loss(dB)/fiberlength(km)$$

$$A(dB/km) = -\frac{10}{L}\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

- The quantity, *A (dB/km)*
  - is called attenuation and it is one of the most important characteristics of an optical fiber

Calculation of total Attenuation

- From the attenuation equation we have

$$P_{out} = P_{in} 10^{\frac{-AL}{10}}$$

- Pre-repeater distance

$$L = \frac{10}{A} \log_{10}\left(\frac{P_{in}}{p_{out}}\right)$$

Problem

- Calculate the maximum transmission distance for a fiber link with an attenuation of 0.5 dB/km if the power launched is 1 mW and receiver sensitivity is 50 μW

Problem

- A certain optical fiber has an attenuation of 0.6 dB/km at 1310 nm and 0.3 dB/km at 1550 nm. Suppose the following two optical signals are launched simultaneously into the fiber:
  a. An optical power of 150 μW at 1310 nm
  b. An optical power of 100 μW at 1550 nm
- What are the power levels in μW of these two signals at
  i. 8 km
  ii. 20 km
- Comment on your answer

THANK YOU

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